

TwinPrompt: Dual-View Time-Series Prompting for Few-Shot Generative Classification

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Abstract—Few-shot time series classification (TSC) remains challenging because, under extremely limited supervision, a model must not only extract stable class-discriminative patterns from a handful of labeled examples, but also develop a decision mechanism that generalizes to unseen classes. Existing methods typically model time series from a single view, making it difficult to jointly capture local-to-global temporal dependencies and holistic waveform morphology. Meanwhile, although pretrained large language models (LLMs) offer strong conditional generative priors, their effective use in TSC is hindered by the representation gap between continuous-valued time series and the hidden space of language models. Moreover, if prediction is still performed with a conventional discriminative classification head, the generative decision-making capability of LLMs—aligned with their pretraining objective—cannot be fully exploited. To address these issues, we propose a unified generative framework for few-shot TSC. Our method first employs a dual-view time-series encoder, where a temporal branch and a visual branch extract complementary dynamic dependencies and morphological structure patterns, respectively. The resulting representations are then aligned to the hidden space of a frozen LLM through alignment pretraining. Finally, we reformulate classification as constrained generation of class-specific tokens, enabling the model to leverage the conditional generation capability of the LLM for few-shot decision-making, while constrained decoding further reduces prediction uncertainty in low-data regimes. Experiments under a strict few-shot protocol on UCR benchmarks show that the proposed method consistently outperforms existing deep learning baselines across multiple shot settings. Ablation studies further confirm the complementarity of the dual-view design, the stable gains and faster convergence brought by alignment pretraining, and the critical role of constrained decoding in generative few-shot classification.

Index Terms—Time-Series Classification, Few-Shot Learning, Generative Classification, Large Language Models, Prompting

I. Introduction

Time series classification (TSC) is a core problem in machine learning, underpinning applications such as

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healthcare monitoring, industrial diagnosis, human activity recognition, and scientific sensing [1]. While deep neural networks have substantially advanced fully supervised TSC, their success typically depends on having sufficiently many labeled sequences [2]. In many practical settings, however, annotations are scarce, costly, or slow to acquire—for example in rare fault diagnosis, patient-specific monitoring, and newly deployed sensing systems. Few-shot TSC therefore remains a practically important and fundamentally difficult regime: the model must infer class-discriminative structure from only a handful of labeled examples per class while remaining robust to large intra-class variation and cross-domain heterogeneity [3].

Existing few-shot TSC methods have mainly explored meta-learning, prototype- or shapelet-based reasoning, and spectral or contrastive representation learning [3]–[5]. In parallel, recent work has begun to connect pretrained large language models (LLMs) to time-series analysis through quantization, projection, prompt-based adaptation, and multimodal interfaces [6]–[8]. Yet these two lines have not fully met where few-shot TSC is hardest. Current LLM-based pipelines often rely on a single representation pathway, even though recent work increasingly suggests that visualized time-series signals can expose morphology-aware cues that are complementary to sequence-space modeling [9], [10]. They also face a persistent modality gap between continuous temporal features and the hidden space of frozen language models [8], [11]. Moreover, final decisions are frequently made either by external discriminative heads or by unconstrained open-vocabulary generation, both of which are only weakly aligned with the autoregressive objective that causal LLMs are pretrained to solve.

We take the view that effective few-shot TSC with frozen causal LLMs is bottlenecked by three coupled issues: representation, alignment, and decision. First, under extreme label scarcity, a single-view encoder may fail to recover all class-discriminative cues from the input sequence. Second, even strong temporal features are of limited use unless they can be mapped into a latent interface that the frozen LLM can reliably exploit. Third, if the prediction mechanism is mismatched with autoregressive next-token generation, the model may underuse the time-series information already encoded in its internal states. Recent evidence that prompt outputs can substantially underestimate the time-series classification capability contained in LLM representations further reinforces this point [12].

Motivated by this perspective, we propose a dual-

view generative framework for few-shot TSC. A patch-based temporal encoder, inspired by patchified time-series Transformers [13], captures local-to-global temporal dependencies directly in sequence space. In parallel, a visual branch renders each time series as a 2D image and extracts morphology-aware representations that emphasize waveform shape, texture, periodic structure, and spatialized anomalies. Rather than collapsing the two views prematurely, we preserve their token-level outputs and concatenate them before projection, so that the downstream model can condition jointly on temporal dynamics and morphological structure.

To make this dual-view evidence usable by a frozen causal LLM, we project the concatenated temporal tokens into the LLM hidden space and treat them as soft prompt tokens inserted into an instruction-style input. We further adopt a stage-wise alignment strategy before few-shot supervision. The first stage learns to produce LLM-compatible time-series representations and reduces the modality gap between temporal features and language-model hidden states. The second stage performs joint optimization of the dual-view encoder, the projection module, and lightweight LoRA adapters while keeping the main LLM backbone frozen. This design makes alignment a primary training target rather than an incidental by-product of few-shot finetuning, which is especially important when only a small support set is available.

Finally, instead of attaching an external classifier, we reformulate few-shot TSC as constrained label-token generation. We introduce class-specific tokens into the tokenizer and train the model to generate the correct label token conditioned on textual instructions and dual-view time-series soft prompts. During inference, decoding is restricted to the candidate class tokens in the current episode together with the end-of-sequence token. This preserves consistency with the causal LLM’s native generation objective while removing irrelevant vocabulary degrees of freedom, thereby yielding more stable decisions under few-shot uncertainty.

Extensive experiments under a strict support/query protocol validate this formulation. Across shot settings, the dual-view model consistently outperforms either single branch alone; stage-wise alignment pretraining accelerates convergence, stabilizes optimization, and improves final accuracy; constrained decoding materially boosts prediction quality; and the overall framework surpasses strong deep-learning baselines at the same shot level. These results suggest that the key to few-shot TSC is not simply introducing a larger pretrained model, but enabling the model to see richer temporal evidence, align that evidence with a frozen LLM in latent space, and decide through a generation mechanism tailored to the episode label set.

In summary, our contributions are threefold: (1) we introduce a dual-view encoder that couples patch-based temporal modeling with morphology-aware visual modeling for few-shot TSC; (2) we present a stage-wise time-series-to-LLM alignment scheme with parameter-efficient adaptation for frozen causal LLMs; and (3) we cast

few-shot TSC as constrained label-token generation and show that dual-view encoding, alignment pretraining, and constrained decoding each make measurable contributions to performance.

II. Related Work

a) Few-shot time series classification.: Few-shot time series classification (TSC) has mainly been studied through meta-learning, metric learning, and prototype-based inductive biases. Narwariya et al. [14] introduced a meta-learning framework that trains residual networks across heterogeneous UCR tasks for rapid adaptation under scarce supervision. Tang et al. [3] proposed Dual Prototypical Shapelet Networks (DPSN), which combine representative class prototypes with discriminative local shapelets to improve both interpretability and data efficiency. Yang et al. [5] further explored spectral relations and introduced the Spectral Propagation Graph Network (SPGN) to propagate class information through spectrum-wise comparisons. More recently, COSCO [4] improved few-shot multivariate TSC from the optimization perspective by combining sharpness-aware minimization with a prototypical loss. These methods demonstrate the importance of transferable priors and structured inductive biases, but they remain essentially discriminative pipelines and mostly operate within a single temporal representation space.

b) LLMs and foundation models for time series.: Recent surveys show that foundation-model research for time series has rapidly expanded from adapting pretrained language models to building native temporal foundation models [15]–[17]. Representative adaptation-based approaches include TEST [18], which aligns time-series embeddings with the LLM embedding space through text-prototype-aware contrastive learning; Time-LLM [7], which reprograms time-series patches into a frozen LLM via text prototypes and prompt-as-prefix; and S²IP-LLM [19], which aligns time-series embeddings with semantic anchors derived from pretrained word embeddings. LLM4TS [11] further adopts a two-stage alignment-and-task fine-tuning strategy, while OpenTSLM [20] moves toward native time-series language models with soft-prompt and cross-attention interfaces. A recent line of work has also begun to study few-shot classification with LLM-based architectures. The closest to our setting is LLMFew [21], which uses a patch-wise temporal encoder, LoRA-tuned decoder, and a conventional classification head for few-shot multivariate TSC. Compared with this line, we focus on dual-view temporal evidence, explicit stage-wise alignment before few-shot adaptation, and generative label-token prediction rather than an external classifier. We are also motivated by recent ablation studies suggesting that naively attaching time-series inputs to pretrained LLMs is insufficient unless the temporal interface is meaningfully designed and aligned [22].

c) Vision-based representations for time series.: Beyond sequence-space modeling, a growing body of work represents time series as images so that waveform

morphology, texture, periodicity, and holistic structural patterns can be captured by vision backbones. Recent surveys identify this as an increasingly important branch of time-series foundation modeling [9]. In the context of multimodal reasoning, Liu et al. [10] show that visualization can substantially improve LLM reasoning over time series. For classification, TiViT [23] demonstrates that pretrained vision transformers applied to imaged time series can yield highly competitive representations and are complementary to time-series foundation models. These results suggest that visualized signals expose discriminative cues that are not fully captured by sequence-only modeling. However, existing visual approaches are typically used as alternatives to temporal encoders or as interfaces for reasoning, rather than being fused at the token level with a temporal branch for few-shot generative classification.

d) Generative classification and constrained decoding.: Casting classification as generation has been widely explored in NLP, most notably in the text-to-text paradigm of T5 [24]. Rather than relying on an external discriminative head, this paradigm formulates prediction as conditional token generation, making the decision process more naturally aligned with the pretrained objective of language models. In parallel, constrained decoding has been studied as a general mechanism for restricting generation to valid outputs under lexical or structural constraints [25], [26]. Inspired by these ideas, we formulate few-shot time series classification as generation of class-specific label tokens conditioned on textual instructions and time-series soft prompts. Instead of attaching a separate classifier, our model performs prediction in the native autoregressive manner of a frozen causal LLM, while restricting decoding to the candidate label space of the current episode further reduces irrelevant output freedom and improves decision stability under low-data uncertainty.

In summary, our method lies at the intersection of these four lines of work: few-shot TSC, LLM-based time-series adaptation, vision-based time-series representation learning, and generative classification with constrained decoding. It differs from prior work by jointly combining dual-view token-level encoding, stage-wise time-series-to-LLM alignment, and episode-wise constrained label-token generation in a unified framework for few-shot TSC.

A. Problem Setup

1) Few-shot episode formulation: We consider few-shot generative time-series classification under a fixed N -way K -shot protocol, where N is the total number of classes in a given dataset. Let the dataset-level label space be $\mathcal{Y} = \{c_1, \dots, c_N\}$. Following the support/query evaluation paradigm, each episode is denoted by $\mathcal{E} = (\mathcal{S}, \mathcal{Q}, \mathcal{Y})$, where $\mathcal{S} = \{(x_i^s, y_i^s)\}_{i=1}^{NK}$ is the support set and $\mathcal{Q} = \{(x_j^q, y_j^q)\}_{j=1}^M$ is the query set. Each sample is a univariate time series $x \in \mathbb{R}^L$ with label $y \in \mathcal{Y}$, where L denotes the sequence length. In each episode, the support set contains K labeled

examples for every class in $\mathcal{Y}$, while the query set is drawn from the same fixed label space. Therefore, unlike the meta-learning setting with episode-specific class subsets, our few-shot formulation keeps the class space fixed within each dataset and focuses on learning to generalize from limited support examples to query samples under the same full label set.

2) Generative prediction target: Instead of using a dataset-specific discriminative classifier head, we associate each class $c_n \in \mathcal{Y}_{\text{epi}}$ with a dedicated label token $\langle c_n \rangle$ and define the valid label-token set as $\mathcal{T}_{\text{epi}} = \{\langle c_1 \rangle, \dots, \langle c_N \rangle\}$. Given a query sequence x and the current episode label set, Duet predicts the correct class by autoregressively generating a token $t \in \mathcal{T}_{\text{epi}}$ conditioned on the instruction prompt and the encoded time-series evidence. Formally, the model estimates $p_\theta(t \mid x, \mathcal{Y}_{\text{epi}})$ and outputs the label token with the highest constrained likelihood. Therefore, the decision process is cast as constrained label-token generation rather than a softmax over a fixed classifier head, which keeps the prediction mechanism aligned with the causal LLM objective.

B. Overview of Duet

1) Architecture overview: As illustrated in Fig. ??, Duet contains three components: a dual-view time-series encoder, an alignment projector, and a constrained generative predictor. For an input time series x , the temporal branch extracts sequential patch tokens $\mathbf{H}^{\text{ts}} = E_{\text{ts}}(x)$, while the visual branch transforms x into a 2D view and produces morphology-aware tokens $\mathbf{H}^{\text{vis}} = E_{\text{vis}}(x)$. The two views are then fused as

$$\mathbf{Z} = \text{Fuse}(\mathbf{H}^{\text{ts}}, \mathbf{H}^{\text{vis}}),$$

where $\text{Fuse}(\cdot)$ denotes either direct token concatenation or an optional aggregation module. The fused representation is mapped into the hidden space of the frozen LLM through an alignment projector,

$$\mathbf{P} = \text{Proj}(\mathbf{Z}).$$

The projected tokens are inserted as soft prompts into an instruction-style input, yielding

$$\mathbf{H} = [\mathbf{E}_{\text{pre}}; \mathbf{E}_{\text{ts-text}}; \mathbf{P}; \mathbf{E}_{\text{post}}],$$

where $\mathbf{E}_{\text{pre}}$, $\mathbf{E}_{\text{ts-text}}$, and $\mathbf{E}_{\text{post}}$ denote the embeddings of the textual prompt segments. The frozen causal LLM then generates the target class token under the episode-specific constraint set $\mathcal{T}_{\text{epi}}$.

2) Training stages overview: The training pipeline consists of two levels. First, we perform alignment pretraining to reduce the representation gap between time-series features and the hidden space of the frozen LLM. At this stage, the dual-view encoder and the projector are optimized while the main LLM backbone remains frozen. Second, we adapt the model to downstream few-shot classification by introducing class-specific label tokens and training the model to generate the correct token for each query sample. This downstream stage is further divided

into a warm-up phase and a joint tuning phase. The warm-up phase updates the newly introduced class-token parameters together with the encoder and projector, which stabilizes the initially random class-token embeddings. The joint phase additionally enables lightweight LoRA adapters for the LLM, allowing limited language-side adaptation without fully unfreezing the backbone. This stage-wise design improves optimization stability and preserves consistency between time-series prompting and autoregressive label-token prediction.

III. Other Resources

See [?], [?], [?], [?], [?] for resources on formatting math into text and additional help in working with L^AT_EX.

IV. Text

For some of the remainder of this sample we will use dummy text to fill out paragraphs rather than use live text that may violate a copyright.

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$$x = \sum_{i=0}^n 2iQ. \quad (1)$$

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V. Some Common Elements

A. Sections and Subsections

Enumeration of section headings is desirable, but not required. When numbered, please be consistent throughout the article, that is, all headings and all levels of section headings in the article should be enumerated. Primary headings are designated with Roman numerals, secondary with capital letters, tertiary with Arabic numbers; and quaternary with lowercase letters. Reference and Acknowledgment headings are unlike all other section headings in text. They are never enumerated. They are simply primary headings without labels, regardless of whether the other headings in the article are enumerated.

B. Citations to the Bibliography

The coding for the citations is made with the L^AT_EX `\cite` command. This will display as: see [?].

For multiple citations code as follows: `\cite{ref1,ref2,ref3}` which will produce [?], [?], [?]. For reference ranges that are not consecutive code as `\cite{ref1,ref2,ref3,ref9}` which will produce [?], [?], [?], [?]

C. Lists

In this section, we will consider three types of lists: simple unnumbered, numbered, and bulleted. There have been many options added to IEEEtran to enhance the creation of lists. If your lists are more complex than those shown below, please refer to the original “IEEEtran_HOWTO.pdf” for additional options.

A plain unnumbered list:

```
bare_jrnl.tex
bare_conf.tex
bare_jrnl_compsoc.tex
bare_conf_compsoc.tex
bare_jrnl_comsoc.tex
```

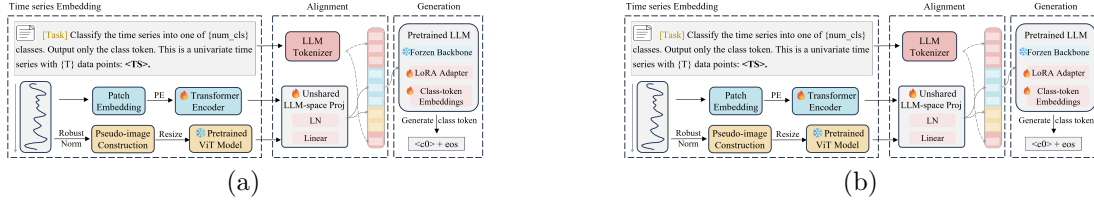


Fig. 1. Dae. Ad quatur autat ut porepel itemoles dolor autem fuga. Bus quia con nessunti as remo di quatus non perum que nimus. (a) Case I. (b) Case II.

A simple numbered list:

- 1) bare_jrnl.tex
- 2) bare_conf.tex
- 3) bare_jrnl_compsoc.tex
- 4) bare_conf_compsoc.tex
- 5) bare_jrnl_comsoc.tex

A simple bulleted list:

- bare_jrnl.tex
- bare_conf.tex
- bare_jrnl_compsoc.tex
- bare_conf_compsoc.tex
- bare_jrnl_comsoc.tex

D. Figures

Fig. 1 is an example of a floating figure using the graphicx package. Note that `\label` must occur AFTER (or within) `\caption`. For figures, `\caption` should occur after the `\includegraphics`.

Fig. 2(a) and 2(b) is an example of a double column floating figure using two subfigures. (The `subfig.sty` package must be loaded for this to work.) The subfigure `\label` commands are set within each subfloat command, and the `\label` for the overall figure must come after `\caption`. `\hfil` is used as a separator to get equal spacing. The combined width of all the parts of the figure should do not exceed the text width or a line break will occur.

Note that often IEEE papers with multi-part figures do not place the labels within the image itself (using the optional argument to `\subfloat[]`), but instead will reference/describe all of them (a), (b), etc., within the main caption. Be aware that for `subfig.sty` to generate the (a), (b), etc., subfigure labels, the optional argument to `\subfloat` must be present. If a subcaption is not desired, leave its contents blank, e.g., `\subfloat[]`.

VI. Tables

Note that, for IEEE-style tables, the `\caption` command should come BEFORE the table. Table captions use title case. Articles (a, an, the), coordinating conjunctions (and, but, for, or, nor), and most short prepositions are lowercase unless they are the first or last word. Table text will default to `\footnotesize` as the IEEE normally uses this smaller font for tables. The `\label` must come after `\caption` as always.

TABLE I
An Example of a Table

One	Two
Three	Four

VII. Algorithms

Algorithms should be numbered and include a short title. They are set off from the text with rules above and below the title and after the last line.

Algorithm 1 Weighted Tanimoto ELM.

TRAIN($\mathbf{X}\mathbf{T}$)

select randomly $W \subset \mathbf{X}$

$N_t \leftarrow |\{i : \mathbf{t}_i = \mathbf{t}\}|$ for $\mathbf{t} = -1, +1$

$B_i \leftarrow \sqrt{\max(N_{-1}, N_{+1})/N_{\mathbf{t}_i}}$ for $i = 1, \dots, N$

$\hat{\mathbf{H}} \leftarrow B \cdot (\mathbf{X}^T \mathbf{W}) / (\|\mathbf{X}\| + \|\mathbf{W}\| - \mathbf{X}^T \mathbf{W})$

$\beta \leftarrow (I/C + \hat{\mathbf{H}}^T \hat{\mathbf{H}})^{-1} (\hat{\mathbf{H}}^T B \cdot \mathbf{T})$

return $\mathbf{W}, \beta$

PREDICT($\mathbf{X}$)

$\mathbf{H} \leftarrow (\mathbf{X}^T \mathbf{W}) / (\|\mathbf{X}\| + \|\mathbf{W}\| - \mathbf{X}^T \mathbf{W})$

return $\text{sign}(\mathbf{H}\beta)$

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VIII. Mathematical Typography and Why It Matters

Typographical conventions for mathematical formulas have been developed to provide uniformity and clarity of presentation across mathematical texts. This enables the readers of those texts to both understand the author's ideas and to grasp new concepts quickly. While software such as L^AT_EX and MathType[®] can produce aesthetically pleasing math when used properly, it is also very easy to misuse the software, potentially resulting in incorrect math display.

IEEE aims to provide authors with the proper guidance on mathematical typesetting style and assist them in writing the best possible article. As such, IEEE has

assembled a set of examples of good and bad mathematical typesetting [?], [?], [?], [?], [?].

Further examples can be found at <http://journals.ieeeauthorcenter.ieee.org/wp-content/uploads/sites/7/IEEE-Math-Typesetting-Guide-for-Latex-Users.pdf>

A. Display Equations

The simple display equation example shown below uses the “equation” environment. To number the equations, use the `\label` macro to create an identifier for the equation. LaTeX will automatically number the equation for you.

$$x = \sum_{i=0}^n 2iQ. \quad (2)$$

is coded as follows:

```
\begin{equation}
\label{deqn_ex1}
x = \sum_{i=0}^n 2iQ.
\end{equation}
```

To reference this equation in the text use the `\ref` macro. Please see (16)

is coded as follows:

Please see (`\ref{deqn_ex1}`)

B. Equation Numbering

Consecutive Numbering: Equations within an article are numbered consecutively from the beginning of the article to the end, i.e., (1), (2), (3), (4), (5), etc. Do not use roman numerals or section numbers for equation numbering.

Appendix Equations: The continuation of consecutively numbered equations is best in the Appendix, but numbering as (A1), (A2), etc., is permissible.

Hyphens and Periods: Hyphens and periods should not be used in equation numbers, i.e., use (1a) rather than (1-a) and (2a) rather than (2.a) for subequations. This should be consistent throughout the article.

C. Multi-Line Equations and Alignment

Here we show several examples of multi-line equations and proper alignments.

A single equation that must break over multiple lines due to length with no specific alignment.

The first line of this example

The second line of this example

The third line of this example (3)

is coded as:

```
\begin{multline}
\text{The first line of this example}\\
\text{The second line of this example}\\
\text{The third line of this example}
\end{multline}
```

A single equation with multiple lines aligned at the = signs

$$a = c + d \quad (4)$$

$$b = e + f \quad (5)$$

is coded as:

```
\begin{align}
a &= c+d \\
b &= e+f
\end{align}
```

The align environment can align on multiple points as shown in the following example:

$$x = y \qquad X = Y \qquad a = bc \quad (6)$$

$$x' = y' \qquad X' = Y' \qquad a' = bz \quad (7)$$

is coded as:

```
\begin{align}
x &= y & X &= Y & a &= bc \\
x' &= y' & X' &= Y' & a' &= bz
\end{align}
```

D. Subnumbering

The `amsmath` package provides a subequations environment to facilitate subnumbering. An example:

$$f = g \quad (8a)$$

$$f' = g' \quad (8b)$$

$$\mathcal{L}f = \mathcal{L}g \quad (8c)$$

is coded as:

```
\begin{subequations}\label{eq:2}
\begin{align}
f&=g \label{eq:2A}\\
f'&=g' \label{eq:2B}\\
\mathcal{L}f &= \mathcal{L}g \label{eq:2c}
\end{align}
\end{subequations}
```

E. Matrices

There are several useful matrix environments that can save you some keystrokes. See the example coding below and the output.

A simple matrix:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (9)$$

is coded as:

```
\begin{equation}
\begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix}
\end{equation}
```

A matrix with parenthesis

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (10)$$

is coded as:

```
\begin{equation}
\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}
\end{equation}
```

A matrix with square brackets

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad (11)$$

is coded as:

```
\begin{equation}
\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}
\end{equation}
```

A matrix with curly braces

$$\begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix} \quad (12)$$

is coded as:

```
\begin{equation}
\begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix}
\end{equation}
```

A matrix with single verticals

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \quad (13)$$

is coded as:

```
\begin{equation}
\begin{vmatrix} a & b \\ c & d \end{vmatrix}
\end{equation}
```

A matrix with double verticals

$$\left\| \begin{matrix} i & 0 \\ 0 & -i \end{matrix} \right\| \quad (14)$$

is coded as:

```
\begin{equation}
\begin{Vmatrix} i & 0 \\ 0 & -i \end{Vmatrix}
\end{equation}
```

F. Arrays

The array environment allows you some options for matrix-like equations. You will have to manually key the fences, but there are other options for alignment of the columns and for setting horizontal and vertical rules. The argument to array controls alignment and placement of vertical rules.

A simple array

$$\left(\begin{array}{cccc} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \quad (15)$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccc}
a+b+c & uv & x-y & 27 \\
a+b & u+v & z & 134
\end{array}
\right)
\end{equation}
```

A slight variation on this to better align the numbers in the last column

$$\left(\begin{array}{cccc} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \quad (16)$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccc}
a+b+c & uv & x-y & 27 \\
a+b & u+v & z & 134
\end{array}
\right)
\end{equation}
```

An array with vertical and horizontal rules

$$\left(\begin{array}{c|cc|c} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \quad (17)$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{c|cc|c}
a+b+c & uv & x-y & 27 \\
a+b & u+v & z & 134
\end{array}
\right)
\end{equation}
```

Note the argument now has the pipe “|” included to indicate the placement of the vertical rules.

G. Cases Structures

Many times cases can be miscoded using the wrong environment, i.e., array. Using the cases environment will save keystrokes (from not having to type the \left\lbracket) and automatically provide the correct column alignment.

$$z_m(t) = \begin{cases} 1, & \text{if } \beta_m(t) \\ 0, & \text{otherwise.} \end{cases}$$

is coded as follows:

```
\begin{equation*}
\{z\_m(t)\} =
\begin{cases}
1, & \{\text{if}\} \{ \beta\_m(t) \} \\
0, & \{\text{otherwise.}\}
\end{cases}
\end{equation*}
```

\end{cases}
\end{equation*}

Note that the “&” is used to mark the tabular alignment. This is important to get proper column alignment. Do not use \quad or other fixed spaces to try and align the columns. Also, note the use of the \text macro for text elements such as “if” and “otherwise.”

H. Function Formatting in Equations

Often, there is an easy way to properly format most common functions. Use of the \ in front of the function name will in most cases, provide the correct formatting. When this does not work, the following example provides a solution using the \text macro:

$$d_R^{KM} = \arg \min_{d_i^{KM}} \{d_1^{KM}, \dots, d_6^{KM}\}.$$

is coded as follows:

```
\begin{equation*}
d_{\text{R}}^{\text{KM}} = \underset{\text{\{d_{\text{1}}^{\text{KM}}\}}}{\text{\{\arg min\}}} \text{\{d_{\text{1}}^{\text{KM}}\}},
\ldots, d_{\text{6}}^{\text{KM}}\}.
\end{equation*}
```

I. Text Acronyms Inside Equations

This example shows where the acronym “MSE” is coded using \text{} to match how it appears in the text.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

```
\begin{equation*}
\text{\text{MSE}} = \frac{\text{\{1\}}{\text{\{n\}}}\sum_{\text{\{i=1\}}^{\text{\{n\}}}}
(\text{\{Y_{\text{i}}\}} - \text{\hat{\text{Y}}_{\text{i}}})^2
\end{equation*}
```

IX. Conclusion

The conclusion goes here.

Acknowledgments

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

Appendix

Proof of the Zonklar Equations

Use \appendix if you have a single appendix: Do not use \section anymore after \appendix, only \section*. If you have multiple appendixes use \appendices then use \section to start each appendix. You must declare a \section before using any \subsection or using \label (\appendices by itself starts a section numbered zero.)

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